QUIZ: Vectors **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

1) A sailboat travels 6.00 km due east, then 3.50 km in a direction 35.0° north of east. Use the component method to find the resultant displacement. Show all work on the grid, and report the magnitude and direction (measured as an angle north of due east) in the final answer box.

1

Final Answer

2) A delivery drone flies 250.0 m due north, then 350.0 m at 60.0° north of east, and finally 200.0 m due east. Use the component method to find the resultant displacement. Show all work on the grid, and report the magnitude and direction (measured as an angle north of due east) in the final answer box.

2

Final Answer

3) A gull flies 30.0 m due east, then 50.0 m at 45.0° north of west, then 20.0 m due north. Use the component method to find the resultant displacement. Show all work on the grid, and report the magnitude and direction (measured as an angle north of due west) in the final answer box.

3

Final Answer

4) A scavenger-hunt team walks four legs: 2.00 km due north, 3.00 km at 25.0° north of east, 1.50 km due west, and 2.50 km at 55.0° north of east. Use the component method to find the resultant displacement. Show all work on the grid, and report the magnitude and direction (measured as an angle north of due east) in the final answer box.

4

Final Answer